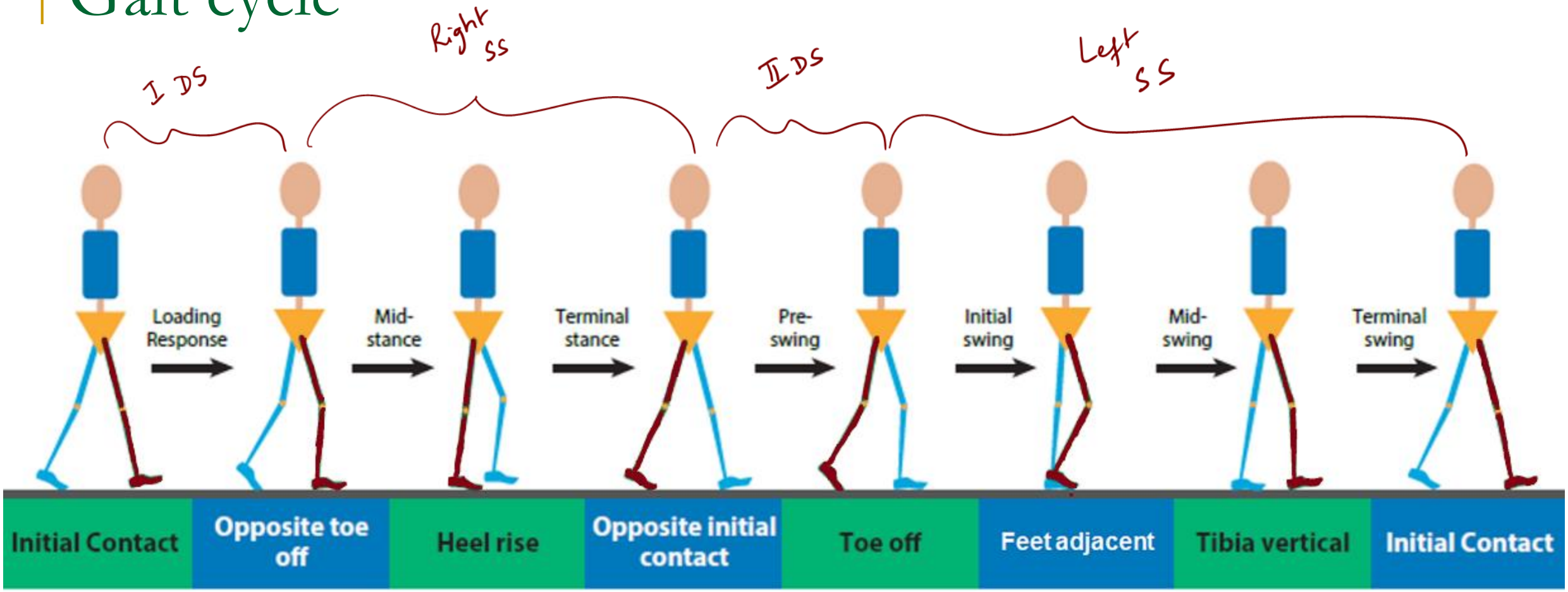
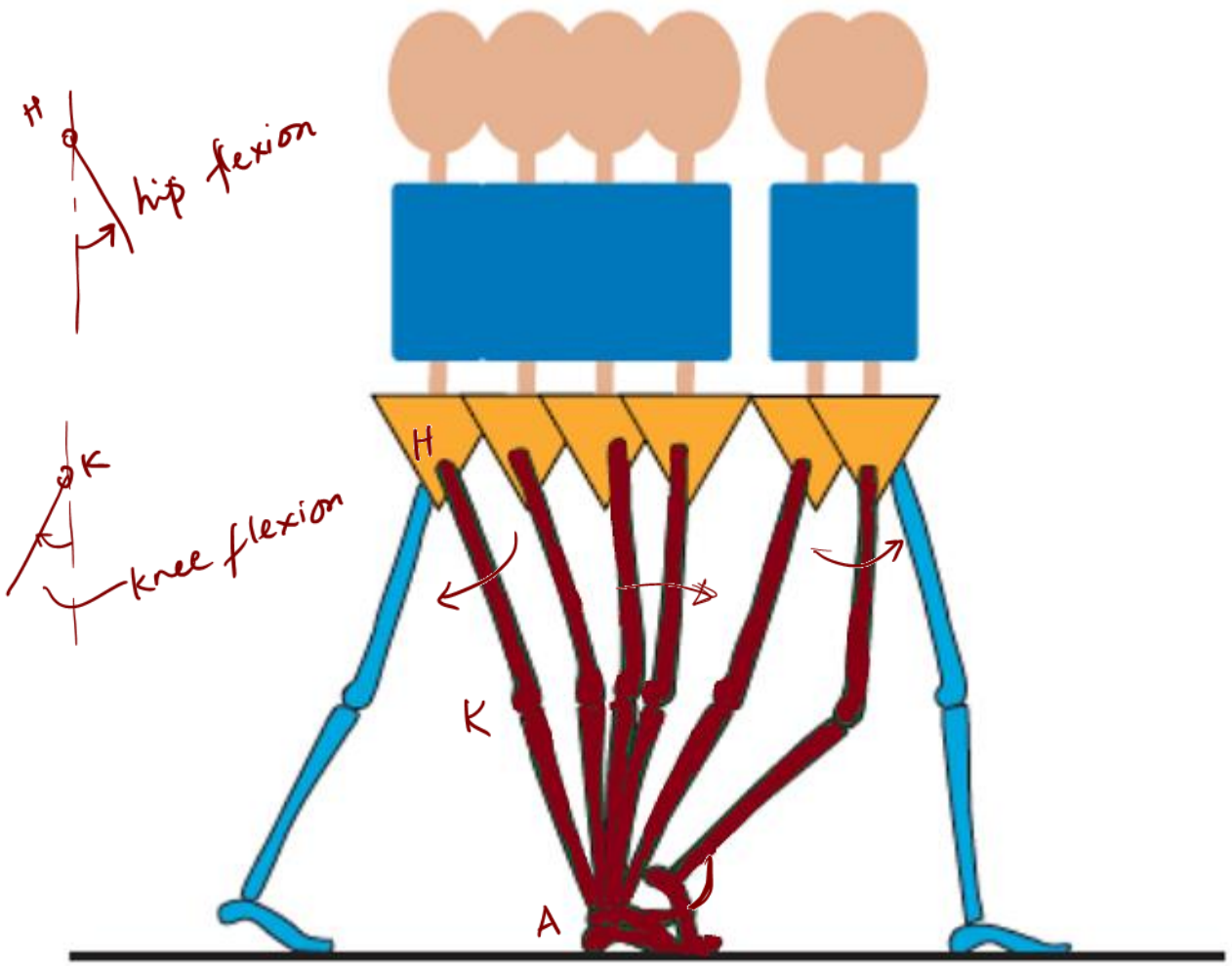
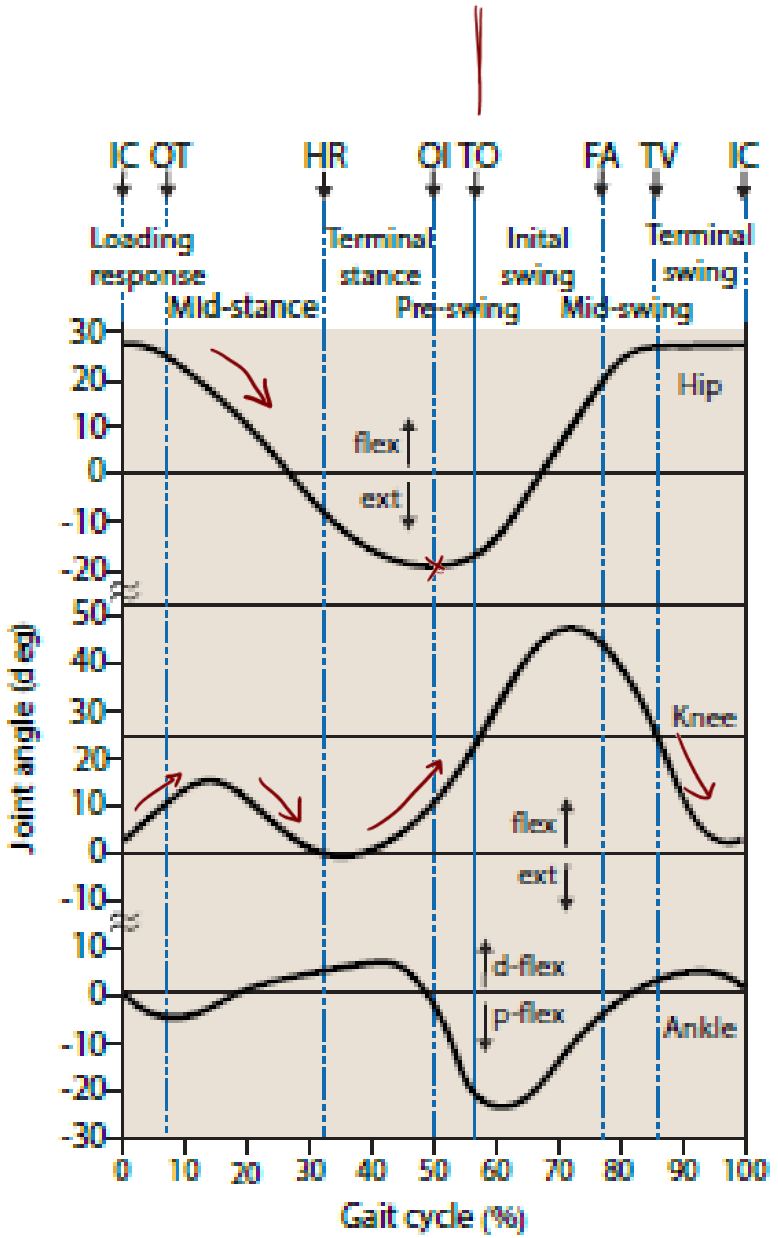
Gait cycle


I DS
Right SS
II DS
Left SS
Loading Response
Mid-stance
Terminal stance
Pre-swing
Initial swing
Mid-swing
Terminal swing
Initial Contact
Opposite toe off
Heel rise
Opposite initial contact
Toe off
Feet adjacent
Tibia vertical
Initial Contact

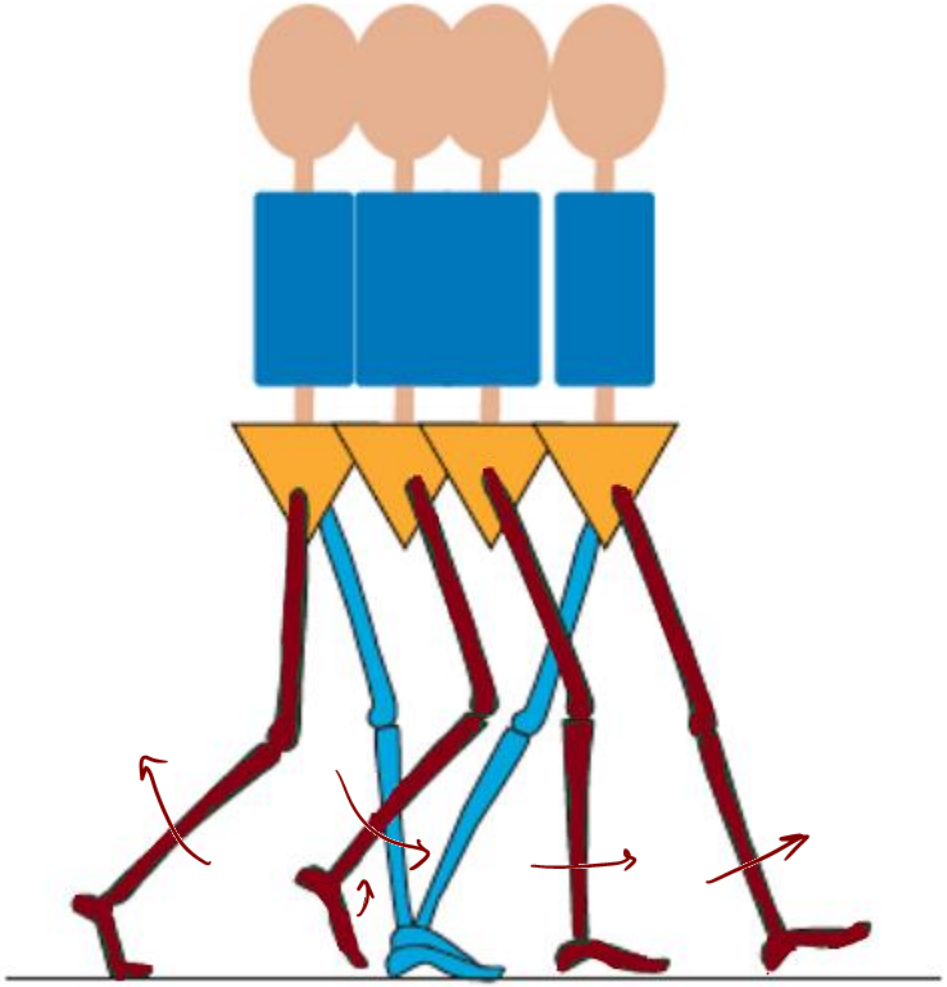
Stance limb during gait

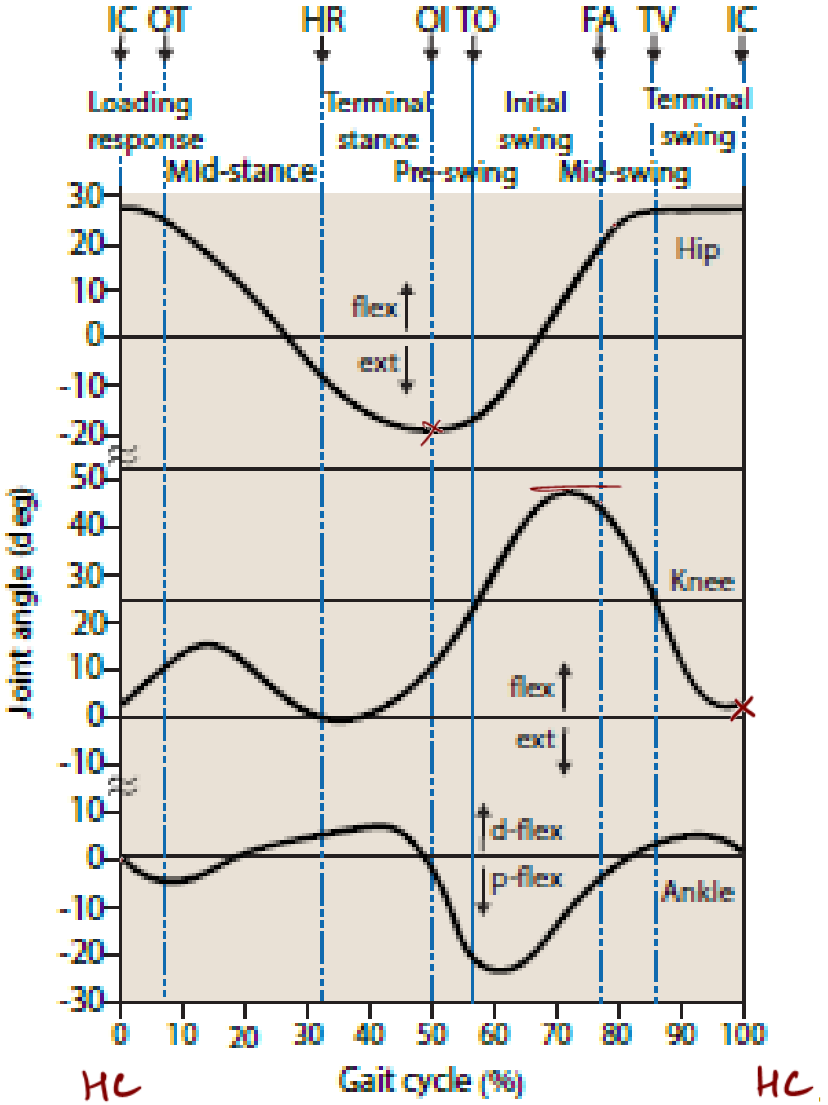




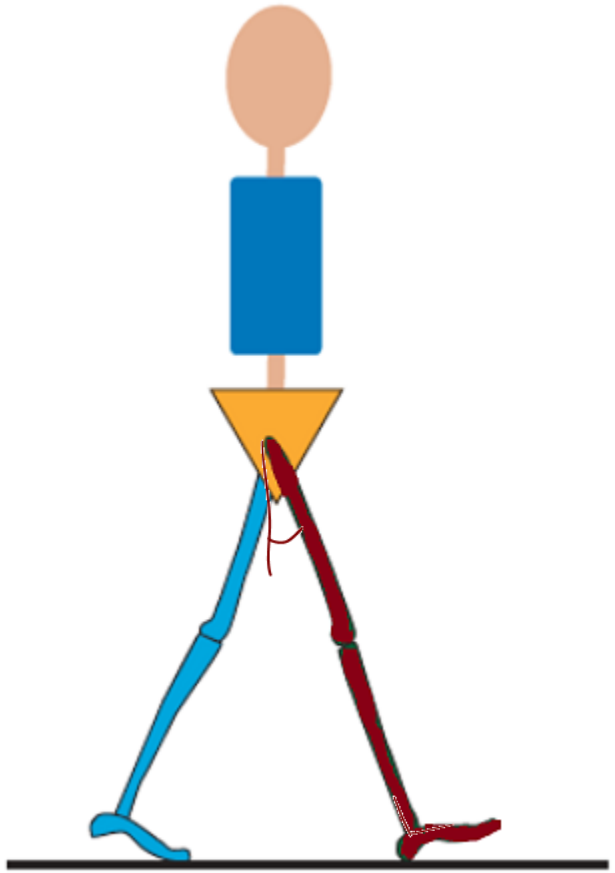
TTK Center for Rehabilitation Research & Device Development (R2D2)
Department of Mechanical Engineering
https://home.iitm.ac.in/r2d2

Swing limb during gait

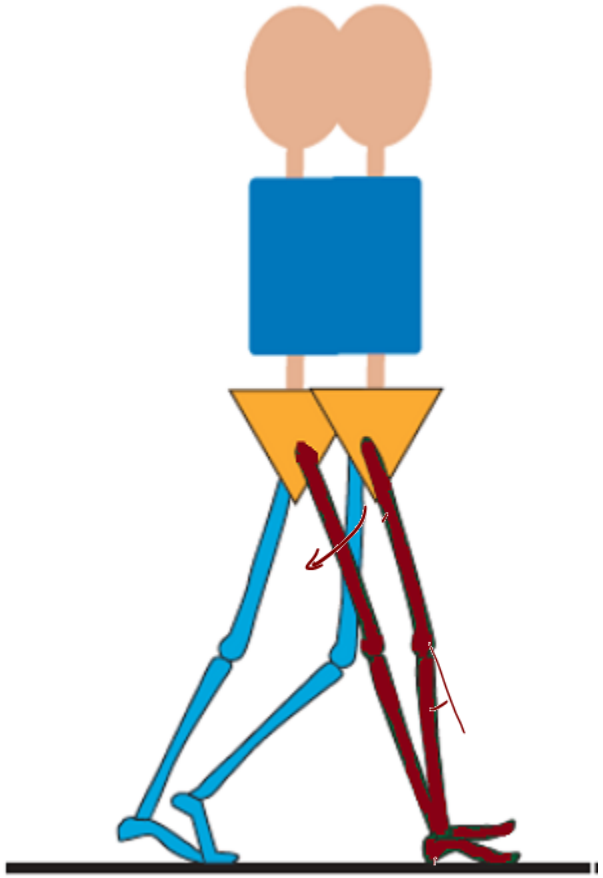




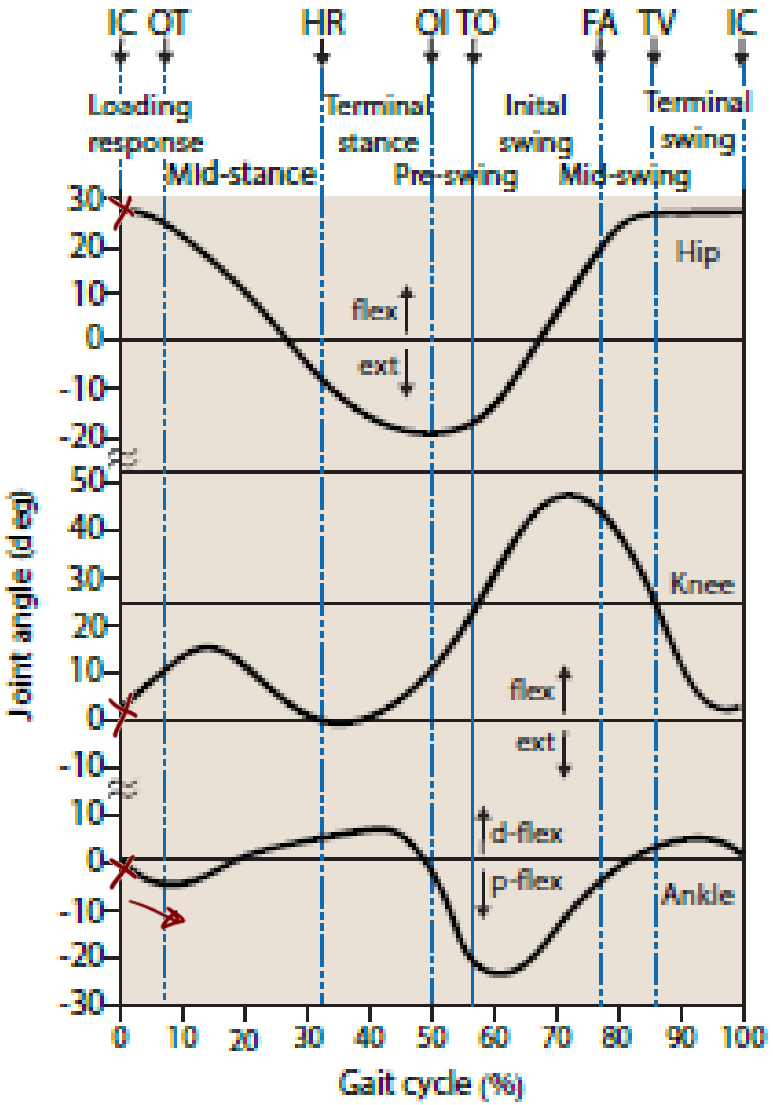
Stance phase



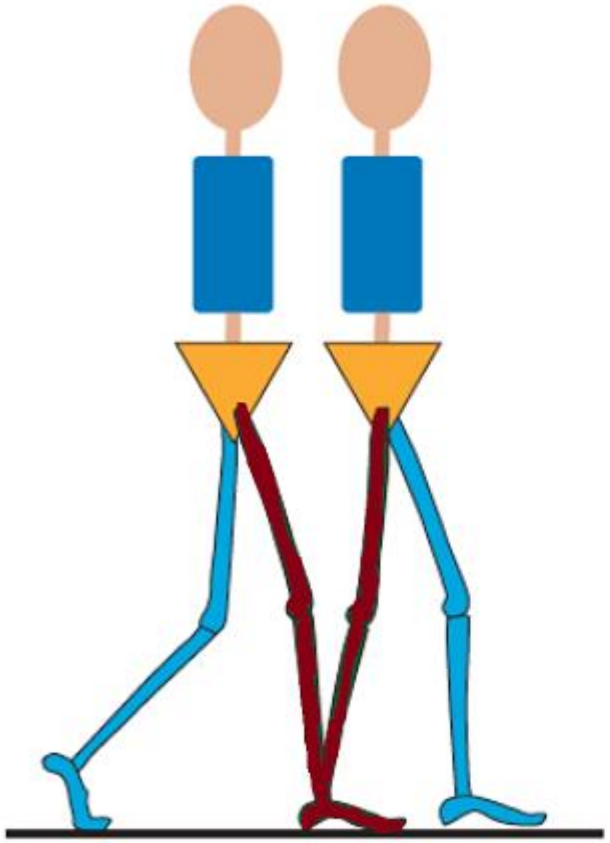
Initial Contact



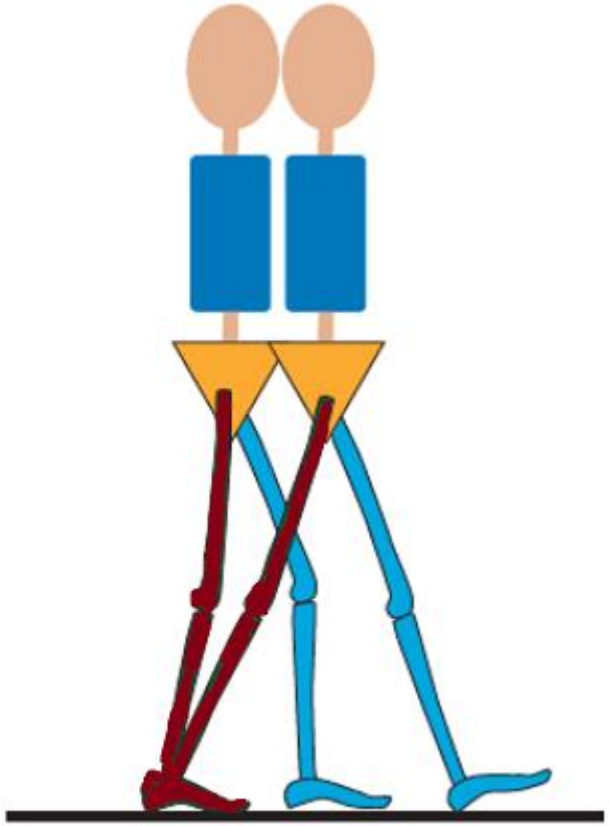
Loading Response



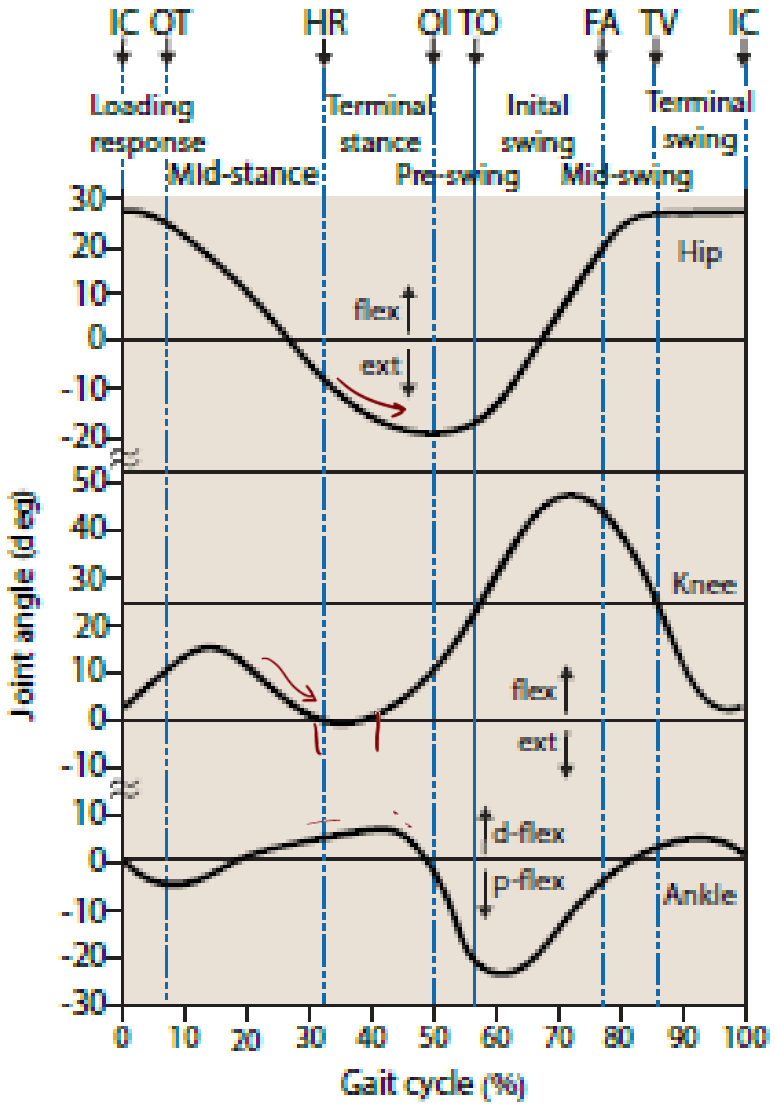
Stance phase



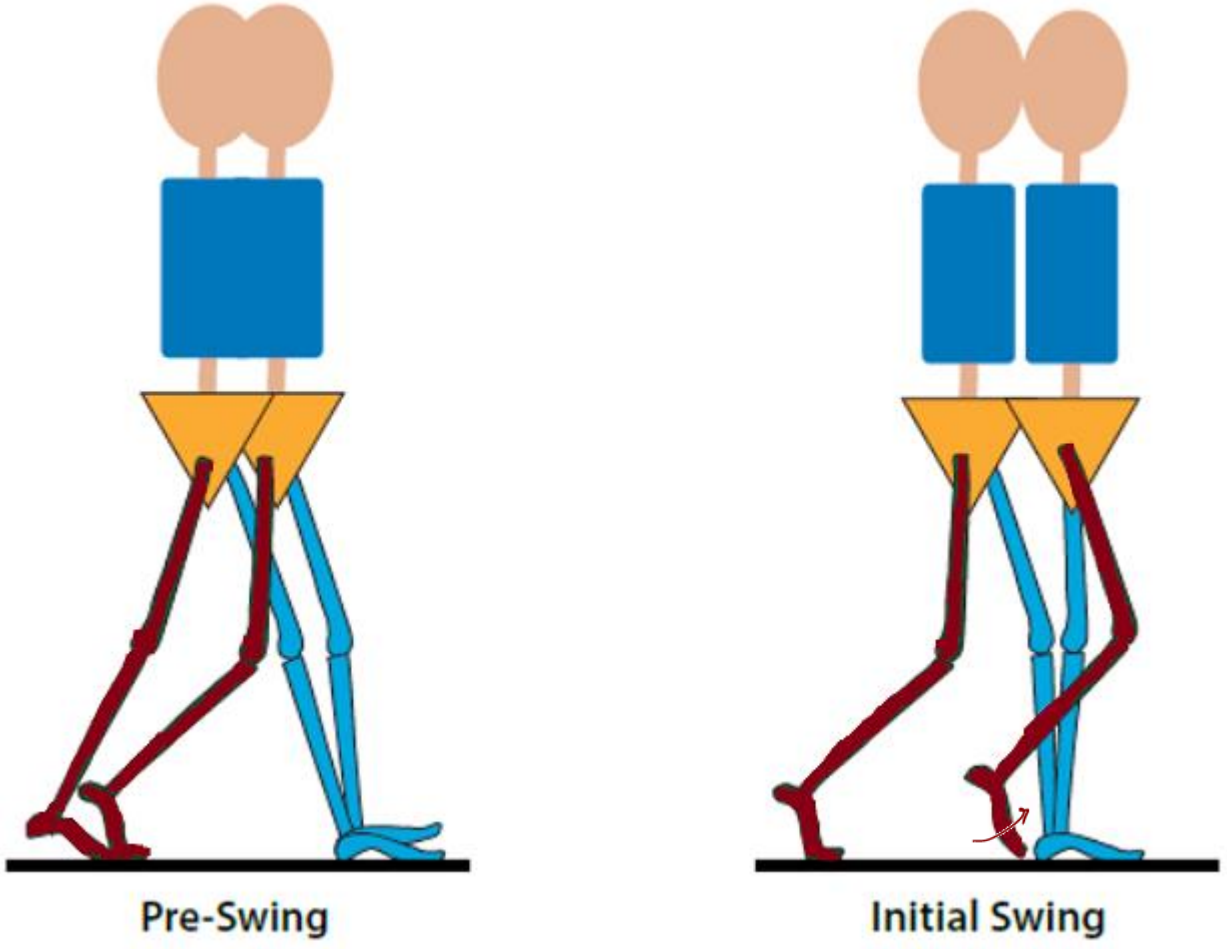
Mid Stance

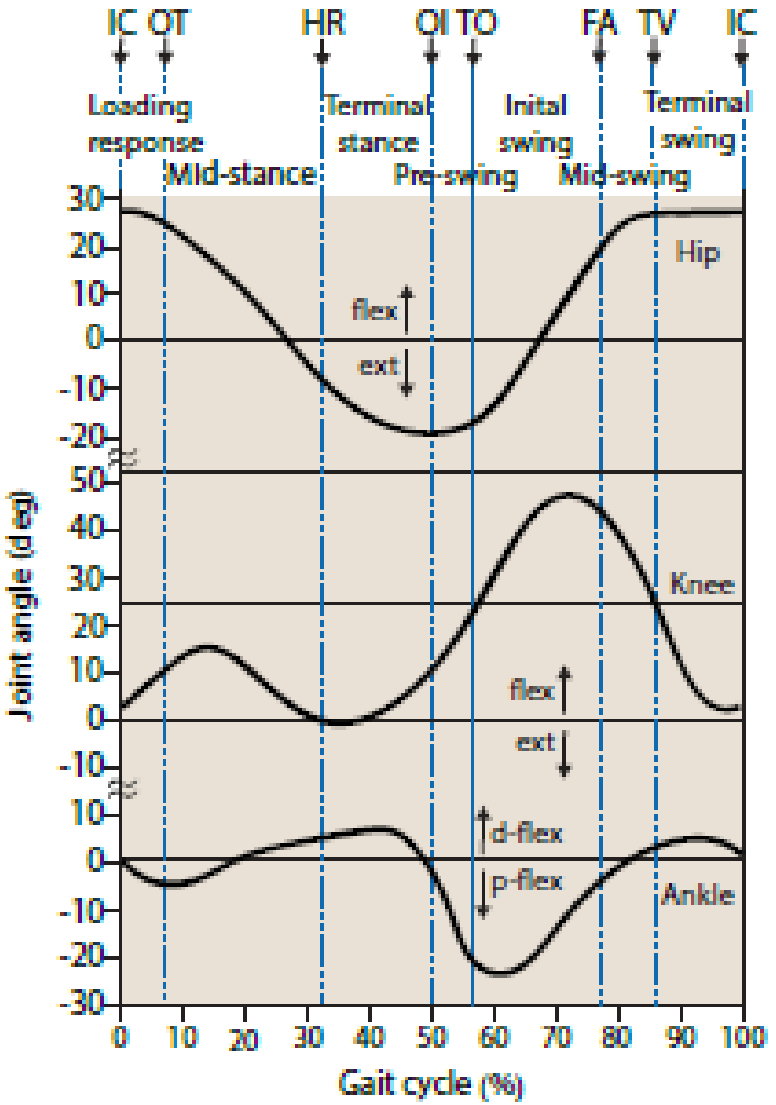


Terminal Stance

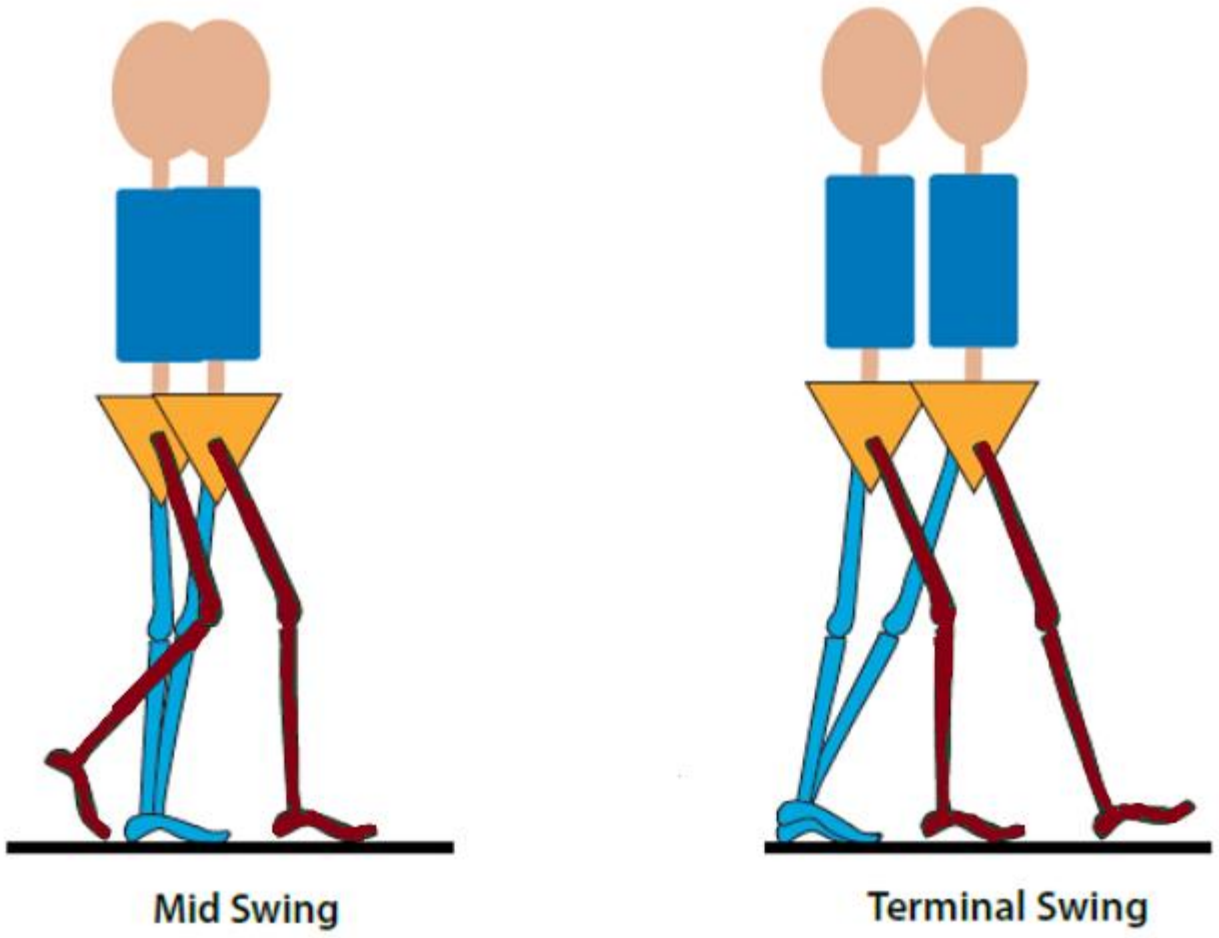


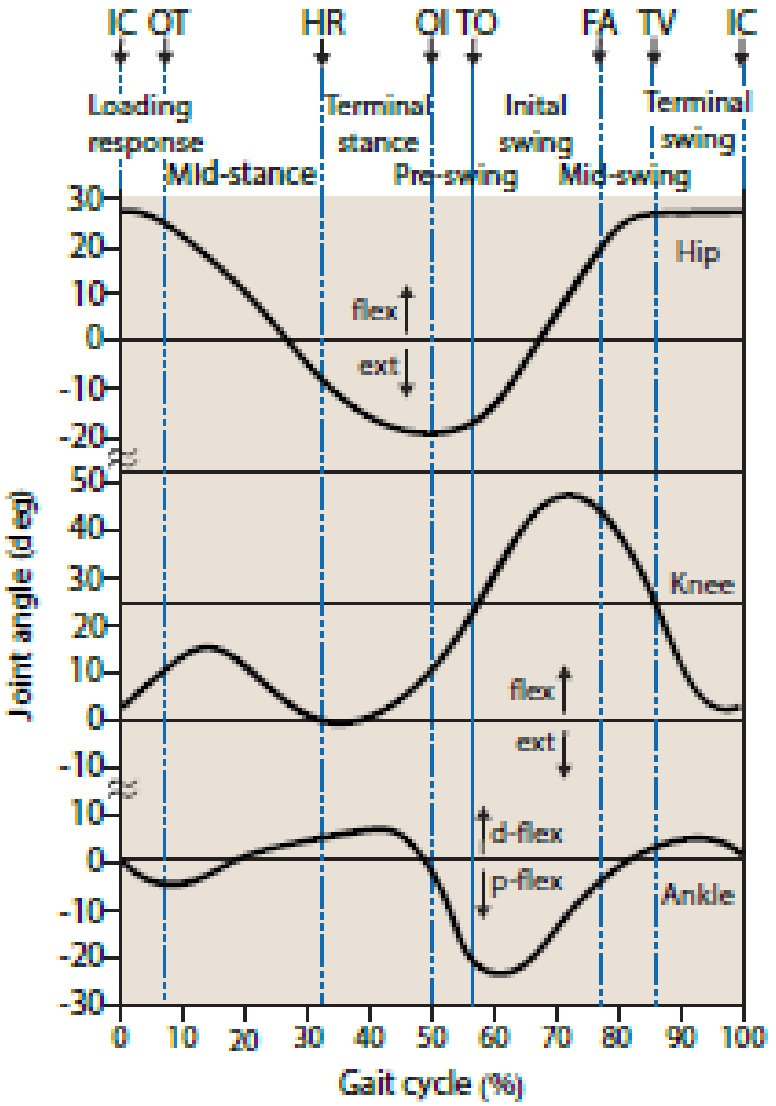
Swing phase


Pre-Swing
Initial Swing

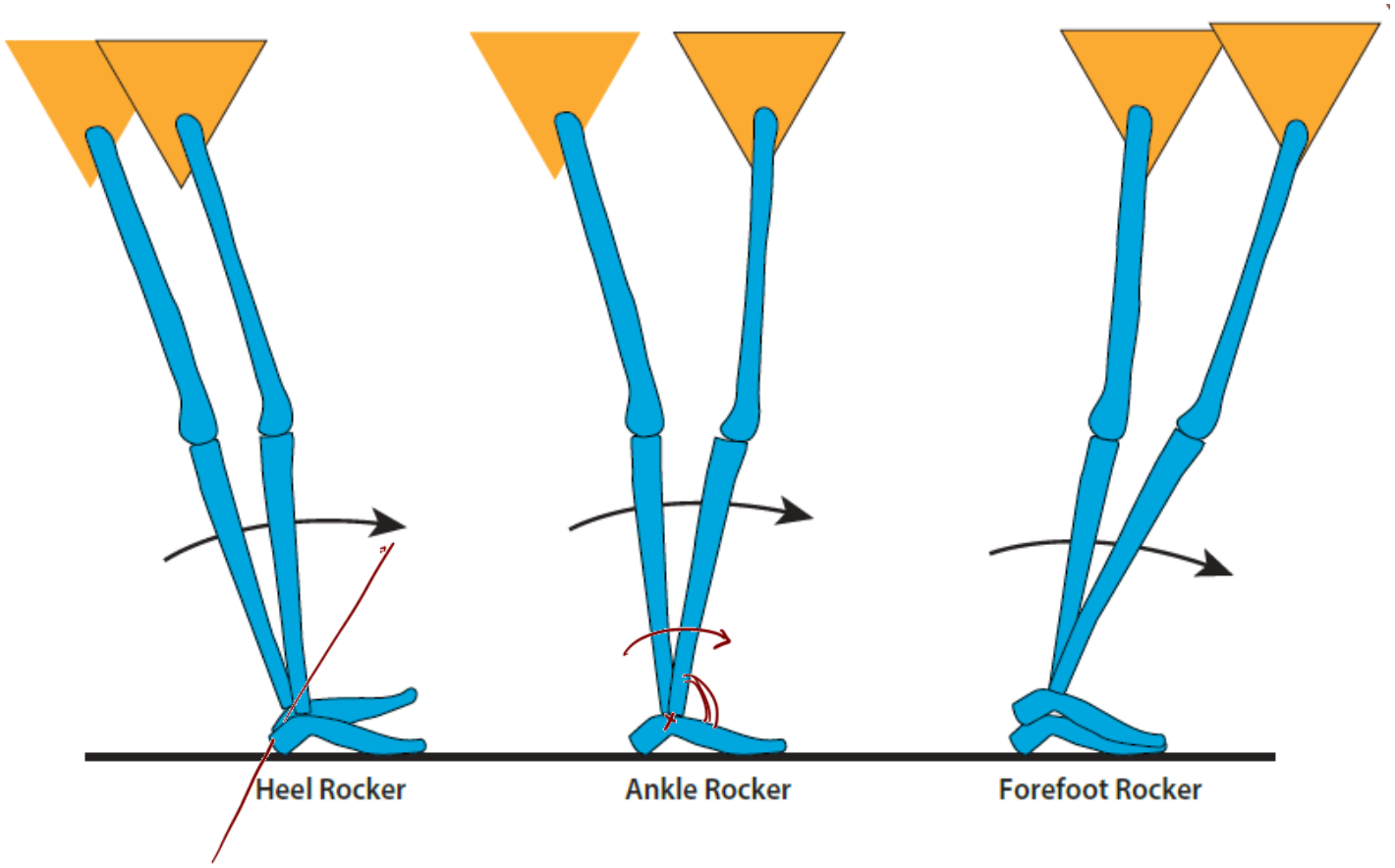

IC
OT
HR
OI
TO
FA
TV
IC
Loading response
Mid-stance
Terminal stance
Pre-swing
Inital swing
Mid-swing
Terminal swing
Hip
flex
ext
Knee
flex
ext
d-flex
p-flex
Ankle
Joint angle (deg)
Gait cycle (%)

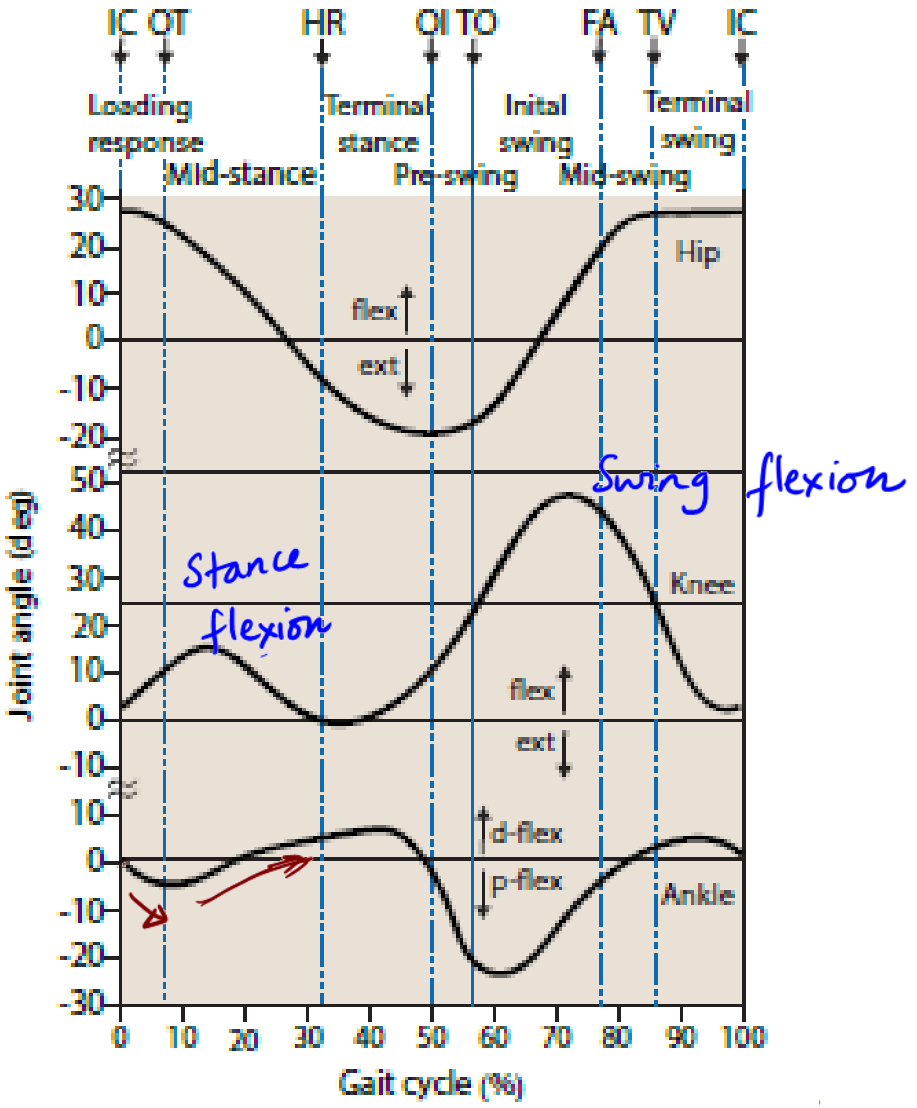
Swing phase


Mid Swing
Terminal Swing

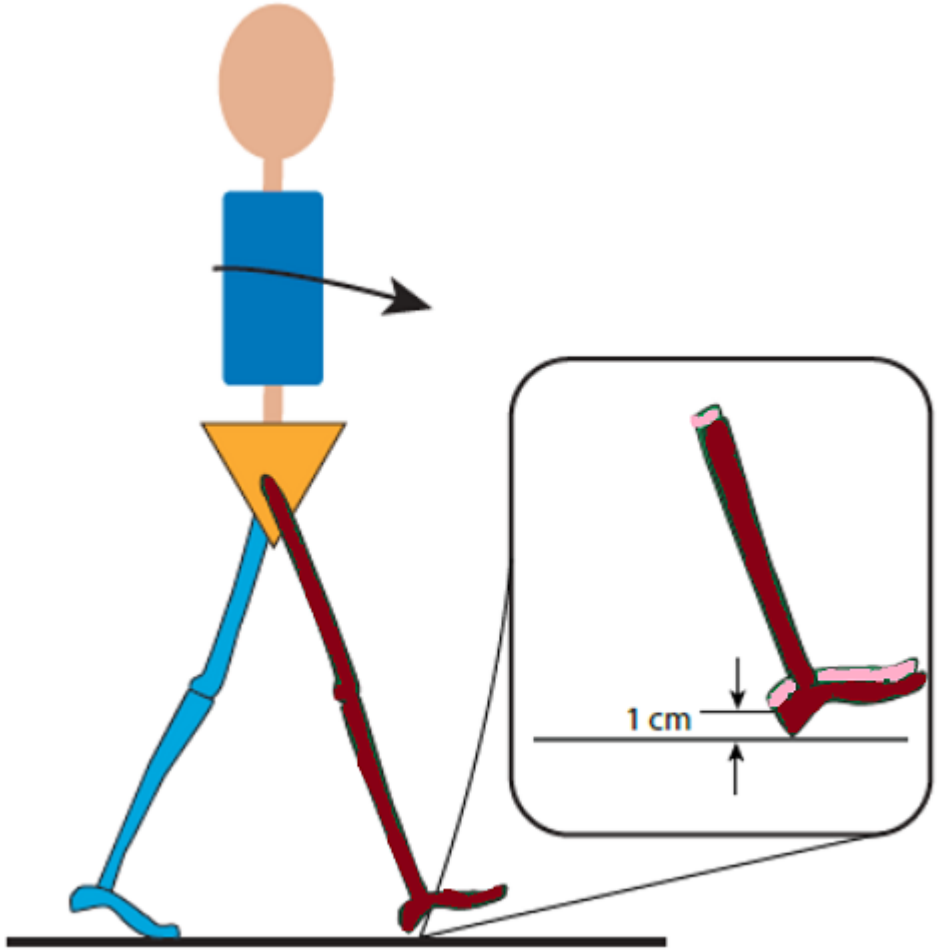

IC
OT
HR
OI
TO
FA
TV
IC
Loading response
Mid-stance
Terminal stance
Pre-swing
Inital swing
Mid-swing
Terminal swing
Hip
flex
ext
Knee
flex
ext
d-flex
p-flex
Ankle
Joint angle (deg)
Gait cycle (%)

Action of the foot

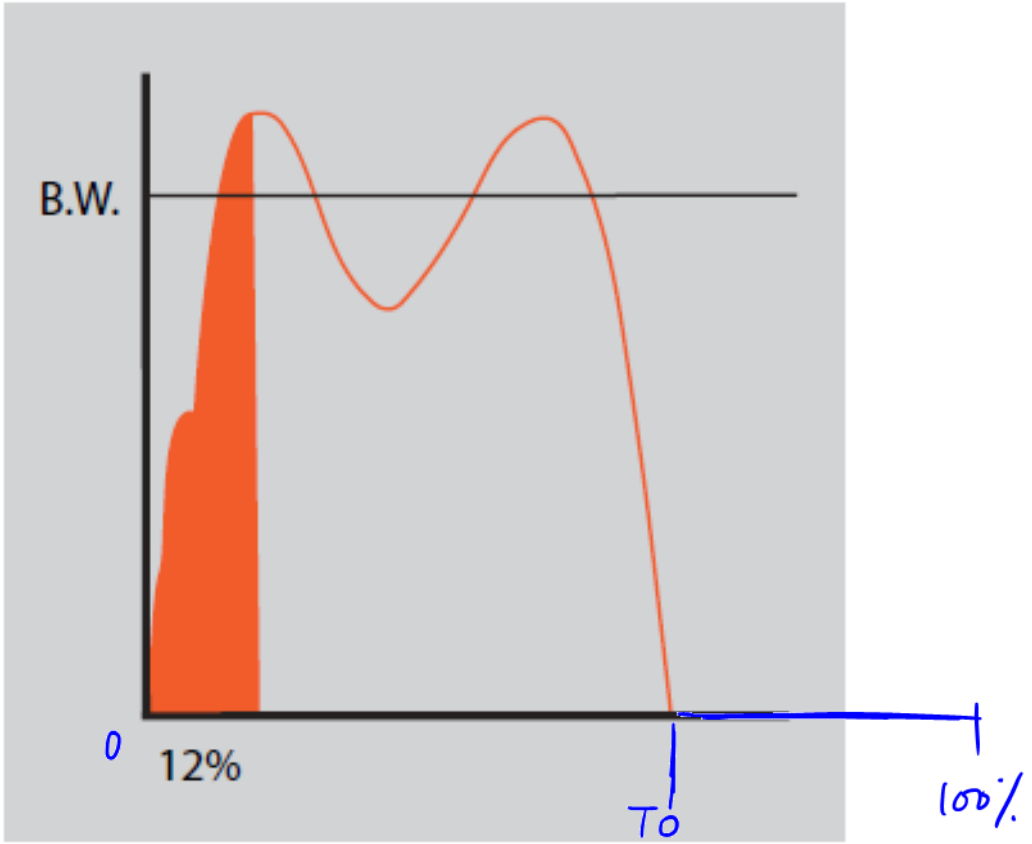




Heel strike and Loading Response

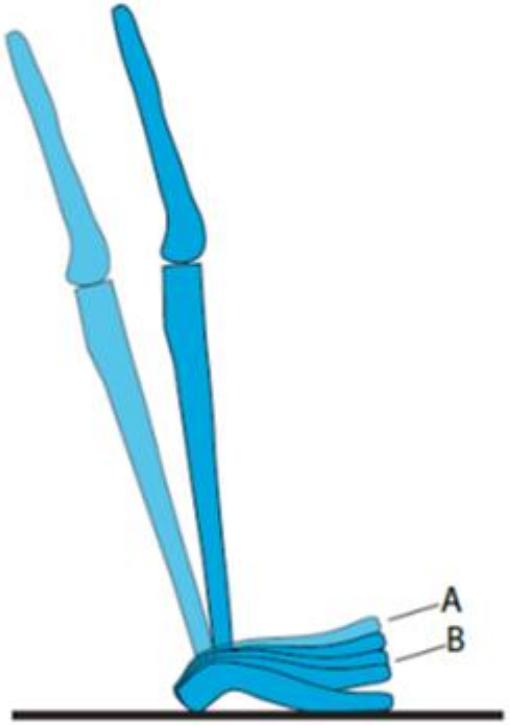


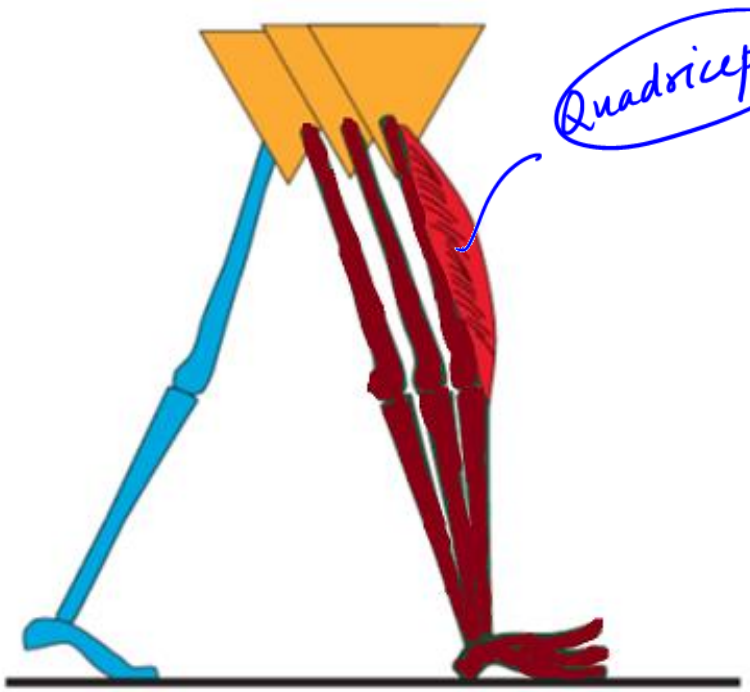
Vertical GRF

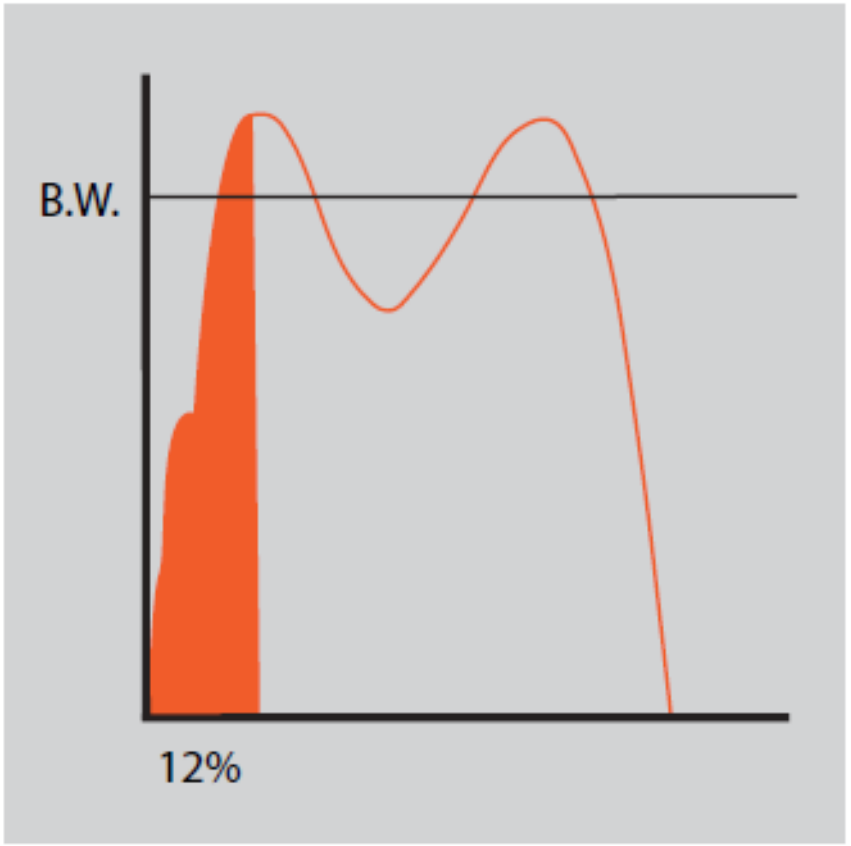


Loading response phase

Shock absorption

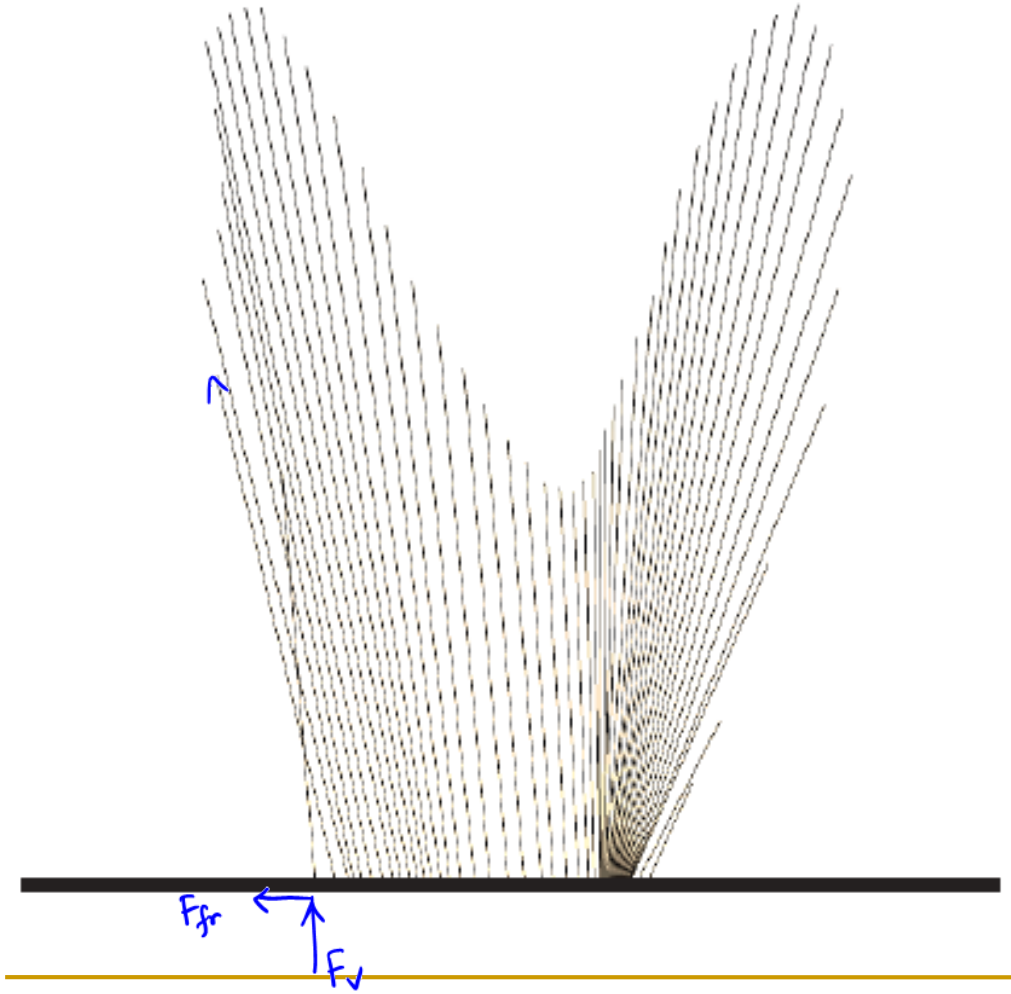




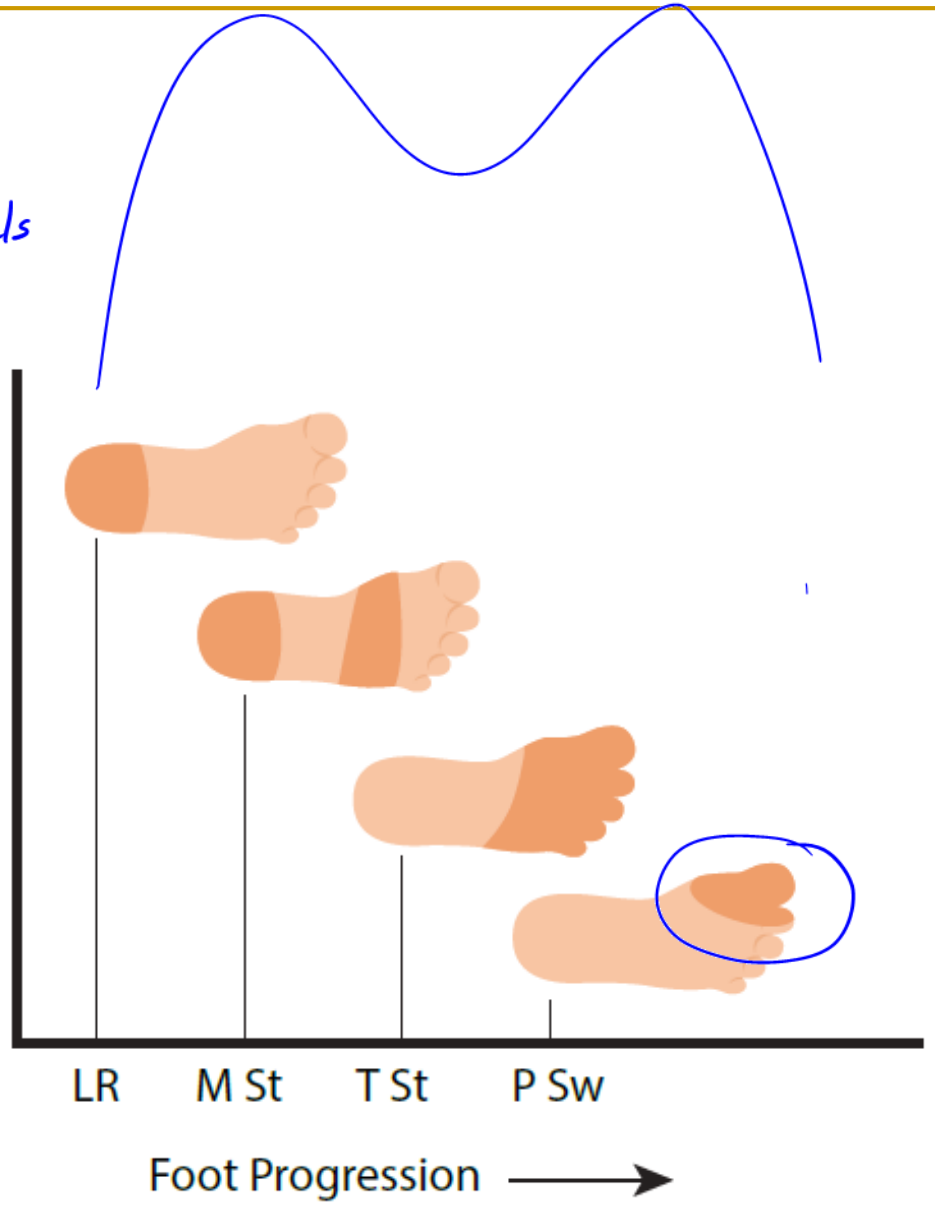


Loading response phase

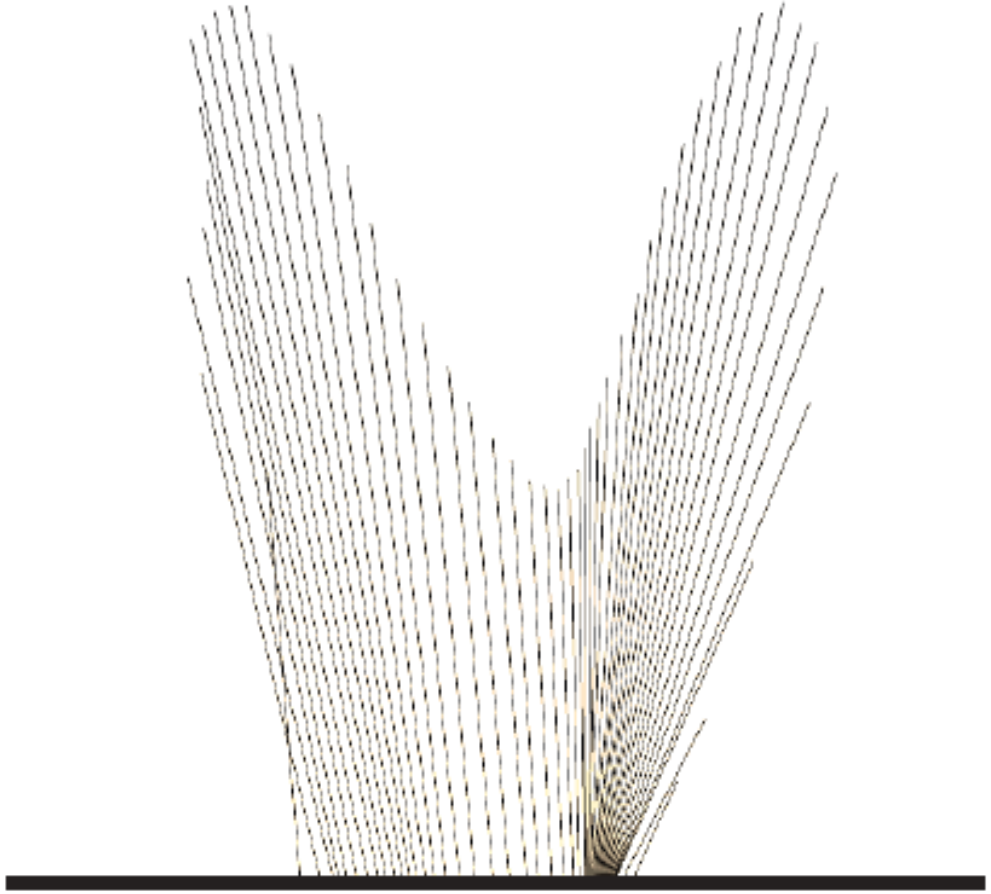
GRF Butterfly Diagram

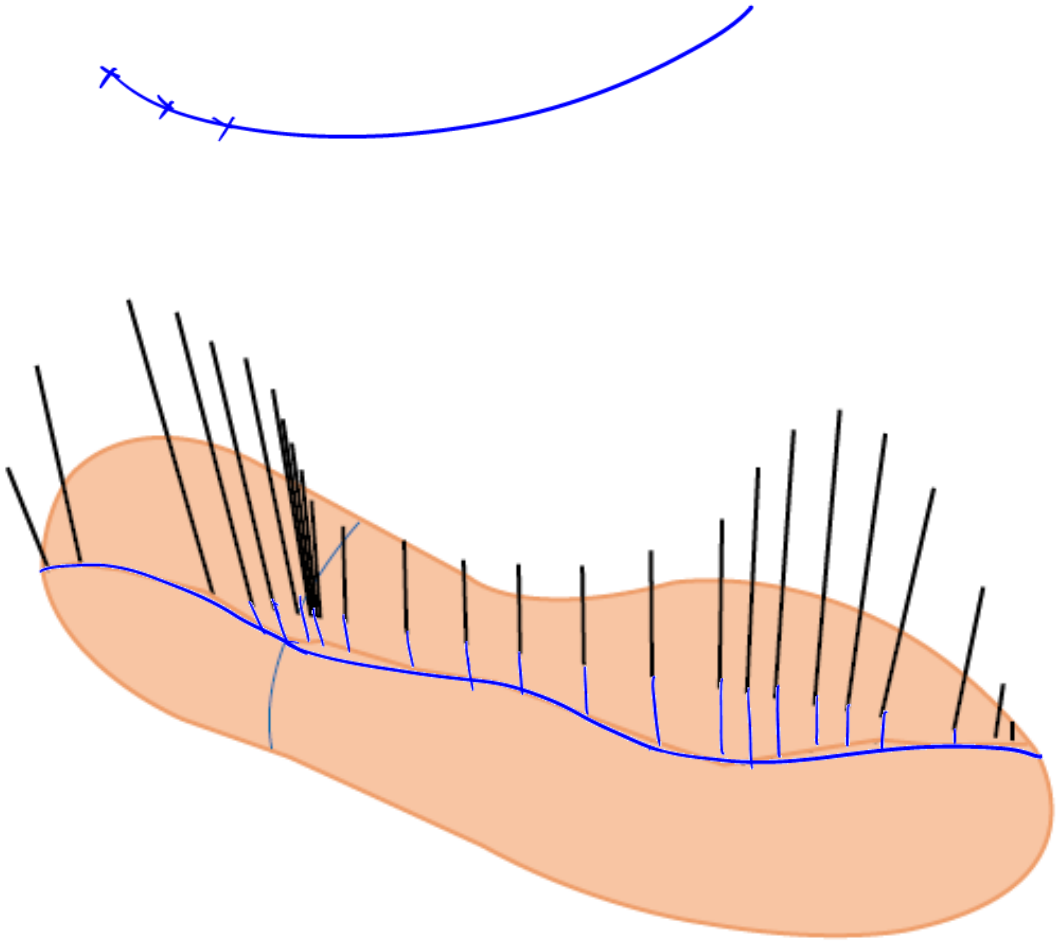


10 ms intervals

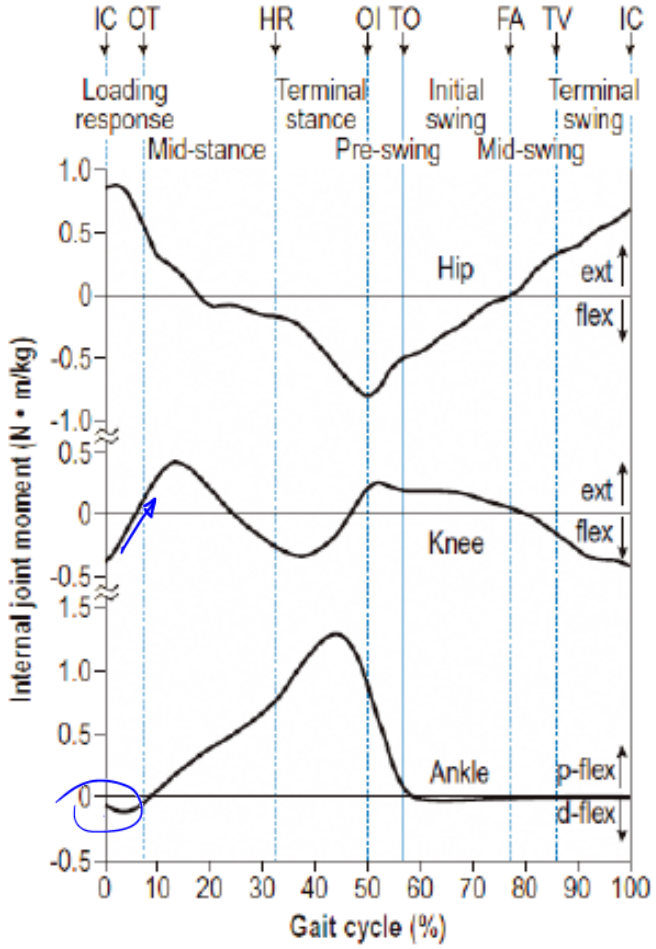


Centre of Pressure





Joint moments



Gait Analysis, Whittle

